System Blueprint

TNPG: PortendedGreatness

Project: Copyrighted Artist

Target ship date: 2026-06-01

Roster:

Name	Email	Primary Role	Secondary Role
Yuhang Pan	yuhangp@nycstudents.net	Project Manager	Game logic/Flask
Andrew Tsai	andrewt194@nycstudents.net	DB Manager	JS Canvas
Zixi Qiao	zixiq@nycstudents.net	Socket	JS Lead
Owen Zeng	owenz20@nycstudents.net	HTML/CSS	JS Sub

Summary

A website hosting a drawing guessing game commonly known as Copyrighted Artists. Upon joining a game lobby, you will be led to a game window where you can draw based on a prompt or based off of what other people have drawn. Then, you will go through the voting phase where you have to determine which one is the real drawing. Score is earned by guessing the real drawing or by fooling others into voting your fake drawings. The site will award you elo points based on winning or losing.

Problem Being Solved

Boredom

Target Users

- Fun-seekers
- Friend-havers
- Unem*loyed-artists

Why This Project Matters

It will help alleviate boredom and train discerning eyes for distinguishing between real and fake drawings

Minimum Viable Product (MVP) Scope

Website with socket lobbies and functioning game

Core Features (Required for Final Submission)

Features that **must** be completed:

1. Socket Lobbies: Users can join different rooms and start separate games; they can also create their own rooms
2. Working Game: Drawing & Voting phases
3. Login DB

Stretch Features (Only if MVP is Complete)

1. Elo System: Gain/lose elo from games. Different elos will be separated into different ranks (i.e. iron, bronze, silver, gold, plat, diamond, etc.)
2. Matchmaking System: Uses the elo system to put you into lobbies of similar player elo.
3. Game history: Be able to review the results of past games, maybe even look at the drawings from those past games.

Explicit Non-Goals

Features intentionally excluded:

- Pay to win
- Unfiltered prompts

Technology Stack

Layer	Selected Tool
Backend Framework	Flask (sockets)
Frontend Framework	tailwind
Database	SQLite
Authentication	Flask sessions
ORM / DB Library	none

Why This Stack Was Chosen

Flask and SQLite for simplicity since all the devos are very familiar with it.

Team Ownership Plan

Each member must own meaningful deliverables.

Team Member	Primary Ownership	Secondary Ownership	Specific Deliverables
Yuhang Pan	Project Manager	Flask + Game Logic	Implement different phases of the game + ensure that they work
Andrew Tsai	DB Manager	JS Canvas	Designing drawing interface of the game using Canvas and send game data to database
Zixi Qiao	Flask Socket	JS Lead	Creating separate lobbies and facilitate displaying updates to users in lobby during the game
Owen Zeng	HTML + CSS	JS Sub	Creating base HTML templates to render pages

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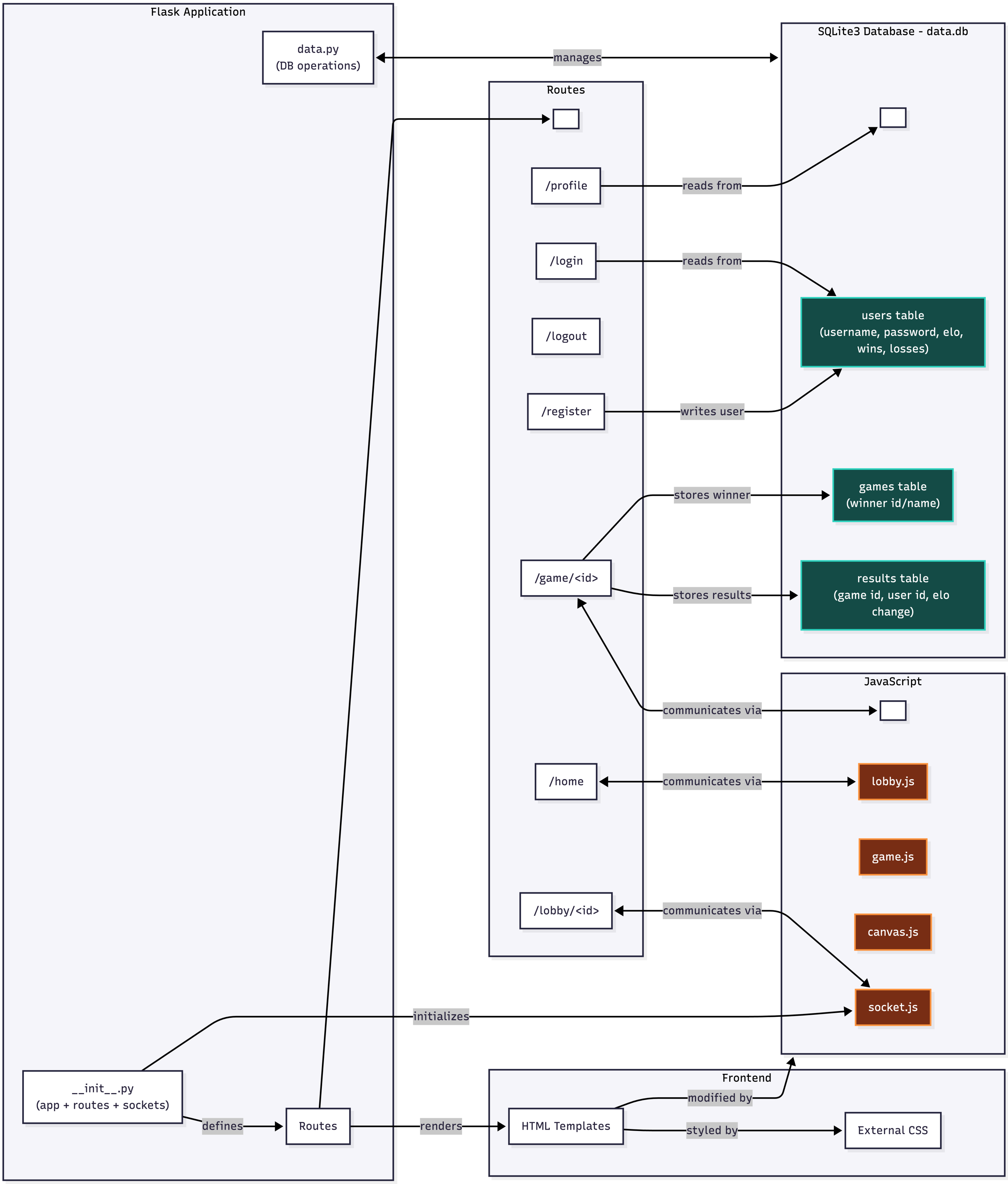
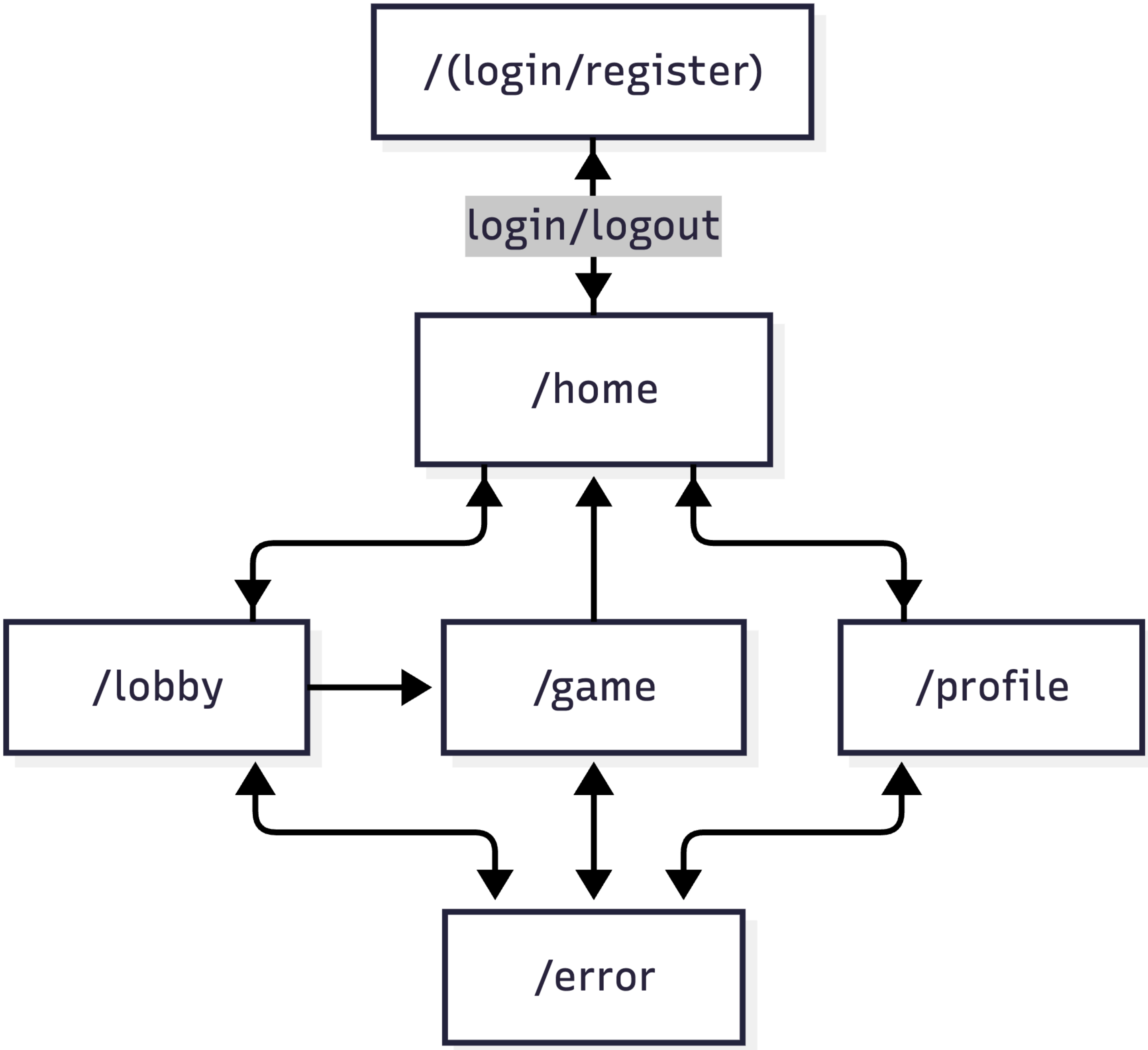
Component map


Flask Application
data.py
(DB operations)
manages
SQLite3 Database - data.db
Routes
/profile
reads from
/login
reads from
users table
(username, password, elo, wins, losses)
/logout
/register
writes user
stores winner
games table
(winner id/name)
/game/<id>
stores results
results table
(game id, user id, elo change)
JavaScript
communicates via
/home
communicates via
lobby.js
game.js
/lobby/<id>
communicates via
canvas.js
initializes
socket.js
Frontend
__init__.py
(app + routes + sockets)
defines
Routes
renders
HTML Templates
modified by
styled by
External CSS

Site map



Key User Stories

Aspiring Artist

As an aspiring artist, I want to train my ability to emulate art styles so that I can become better at drawing. At the same time, I can improve my ability to discern fake/copied arts.

Friend-Havers

As a person with many friends, I want to play a party game so that we can all have fun.

Lone Wolf

As a person with no friends, I want to obsessively play a multiplayer game so that I can beat everyone else and prove my own self-worth.

Database Design

USERS			
INTEGER	id	PK	Auto-increment
TEXT	name		Unique
TEXT	password		For authentication
REAL	elo		
DATE	created_at		
INTEGER	games_won		
INTEGER	games_played		
INTEGER	total_placement		

GAMES			
INTEGER	id	PK	Auto-increment
INTEGER	winner_id	FK	USERS(id)
DATE	timestamp		

RESULTS			
INTEGER	id	PK	Auto-increment
INTEGER	game_id	FK	GAMES(id)
INTEGER	user_id	FK	USERS(id)
REAL	elo_change		

Testing Plan

- First we will check that data storage works properly and authentication is secure by preventing sql injection using `?`
- Then we will make sure all the routes to the web pages work.
- Create lobbies to test socket connections and test sending images drawn on JS canvas
- Test game logic independently by creating a testable lobby connecting all the users to it
- Play through a game and check that user data is updated properly
- Work on extra functions and features

Timeline

Week 1 Goals:

- Authentication
- Working lobbies
- Basic site infrastructure

Week 2 Goals:

- Game logic completed (timer, scoring points)
- Database complete, saving user stats
- JS Canvas working

Week 3 Goals:

- Full game completed (all phases: drawing, voting)
- Using user elo for matchmaking system
- Profile complete
- CSS to make pages look nice

Internal Deadlines:

- Figure out JS canvas and set it up by 5/15 (done)
- Flask sockets for lobbies by 5/18 (done)
- Add start button for lobby host that directs players to the game page by 5/19 (done)
- Set up different phases of the game:
  - Phase 1: all players are given a prompt and 60s to draw and save the sketch by 5/21
  - Phase 2: given an original sketch to copy and save the copied sketch by 5/22
  - Phase 3: go through each drawn prompt, display the 3 sketches and let users vote by 5/25
  - Finally: sum the scores and display leaderboard by 5/28 for live demo

Completion Criteria (*a.k.a.* "Definition of 'Done'")

Project is considered complete when all of the following are true:

1. Socket/lobby system working
2. Game logic + graphics completed
3. User profile data and db secured

Open Questions

Implement really extra features like being able to save certain drawings?

Appendix

Based off of this Roblox game: https://www.roblox.com/games/4353458311/Copyrighted-Artists